BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Hexadecimal

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why does 11111111 plus 00000001 overflow eight bits?

2. A has code 65. Capital-letter codes go in order. What is D's code?
   A) 65
   B) 68
   C) 67
   D) 69

3. "Unicode supports fewer characters than standard ASCII." Correct this. Explain why.

4. A program calculates an area and shows a menu in several places. Correct both claims. Explain why. (a) A function cannot return a value. (b) A procedure must return a value.

Today's objectives

Recall the essential ideas of hexadecimal. Apply a clearly modelled method.

Key vocabulary - Read, cover, recall

Hexadecimal
Base-16 representation using 0-9 and A-F.

Nibble
A group of four bits.

Most significant digit
The digit with the highest positional weight.

Denary
Base-10 representation.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Hex uses base 16

Hexadecimal digits are 0-9 and A-F, where A=10 through F=15.

Four bits per digit

One hexadecimal digit represents exactly four binary bits (one nibble).

Binary to hexadecimal

Split an eight-bit value into two four-bit groups and convert each group to one hex digit.

Hexadecimal to denary

Multiply each hex digit by its positional value, 16 and 1, then add the results.

I Do - Worked model

Convert 10101111 from binary into hexadecimal and then denary. [5 marks]

- Group four bits
- 1010 = 10 = A and 1111 = 15 = F.
- Write AF in hexadecimal.
- In denary, A x 16+F = 10 x 16+15 = 175. [5]

We Do - Guided practice

1. Convert binary 11001010 to hexadecimal. [3]

2. Convert 3E hexadecimal into binary. [3]

3. Convert 2F hexadecimal to denary. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the denary calculation for hexadecimal 2B.

```
first = 2
second = 11
value = ...
print(value)
```

You Do - Independent exam questions

Q2. Convert binary 10010110 to hexadecimal. [3]

Q3. Convert hexadecimal B4 into eight-bit binary. [3]

Q4. Convert hexadecimal 7D to denary. [4]

Exit Ticket - True or False?

A trace table can help check how values change during a program.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. The true sum is 256, requiring nine bits; only eight stored bits would contain 00000000.

2. B - 68

3. Unicode supports far more characters than standard ASCII.

4. (a) A function returns a value, such as an area. (b) A procedure performs a task, such as showing a menu, without returning a value. Use arguments and return values to avoid unwanted changes to global data.

I Do - Explain the model

Group four bits

1010 = 10 = A and 1111 = 15 = F.

Write AF in hexadecimal.

In denary, A x 16+F = 10 x 16+15 = 175. [5]

We Do - Model answers

1. 1100 = C [1], 1010 = A [1], so CA [1].

2. 3 = 0011 [1] and E = 1110 [1], so 00111110 [1].

3. 2 x 16 [1] + 15 x 1 [1] = 32+15 [1] = 47 [1].

Code First - Answer

```
first = 2
second = 11
value = first * 16 + second
print(value)
```

Expected output: 43

You Do - Self-mark

Q2: 1001=9 [1], 0110=6 [1], combined hex 96 [1].

Q3: B=1011 [1], 4=0100 [1], so 10110100 [1].

Q4: 7 x 16 [1] +13 [1] =112+13 [1] =125 [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.